SiLabs Intraoperative Clinical Sentinel Architecture

Data Preprocessing, Base 1D-CNN Models & Tree + CNN Meta-Ensemble

EXECUTIVE SUMMARY: This document details the complete technical architecture for the Silicon Labs Intraoperative Patient Triage & Adverse Event Prediction Engine. Operating at a strict 5-second stride, the system predicts 10-minute future onset of **Hypotension (MAP < 65 mmHg)**, **Hypoxia (SpO2 < 90%)**, and **Tachycardia (HR > 100 bpm)**. Trained on **3,765 perioperative patient records** (VitalDB dataset), the solution combines multi-stage biosignal artifact rejection, 6 base predictive models, and a Meta-Ensemble Neural Network classifier.

1.0 CLINICALLY VALIDATED DATA PREPROCESSING PIPELINE & BIOSIGNAL CHARTS

1.1 Raw vs Cleaned Biosignal Waveforms & Hemodynamic Hierarchy

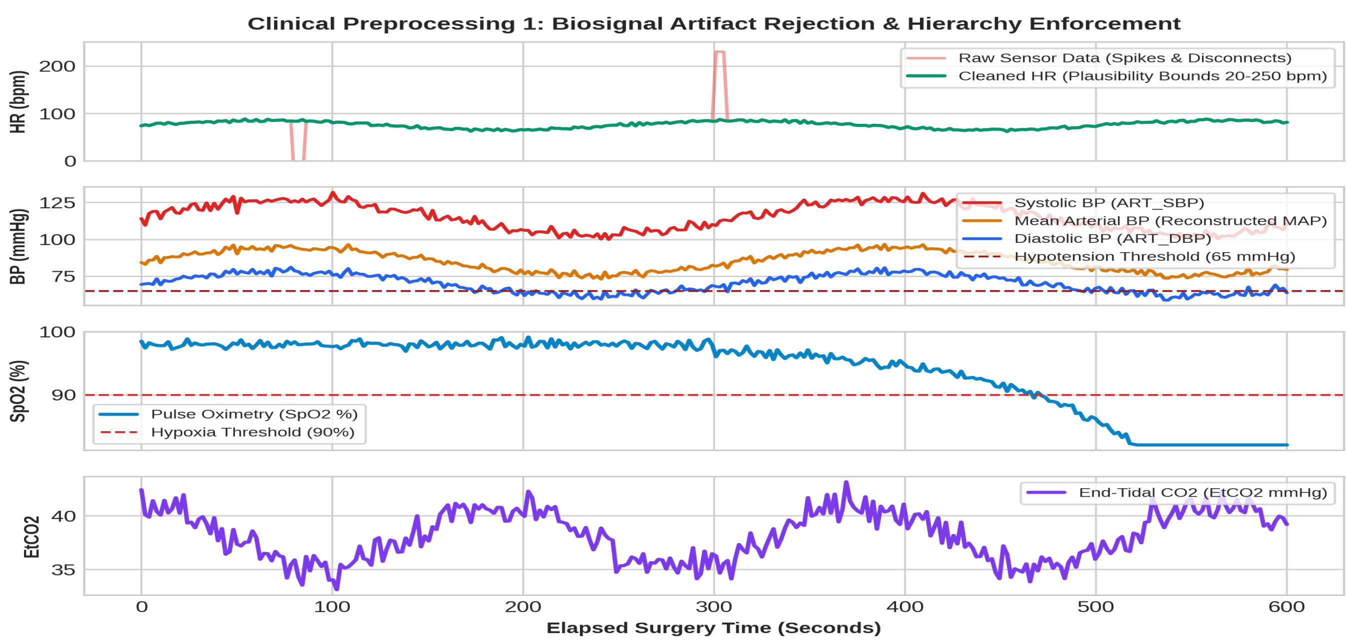

Figure 1: Biosignal Preprocessing Pipeline — Raw Sensor Artifact Rejection & Hemodynamic Hierarchy Enforcement (data_preprocessing.ipynb)

EXPLANATION OF FIGURE 1 (BIOSIGNAL PREPROCESSING & HIERARCHY):

- **HR Artifact Rejection (Panel 1):** Raw continuous telemetry contains sensor disconnect codes (-49.0, -32768) and non-physiological spikes (> 250 bpm). Plausibility bounds (20-250 bpm) isolate true cardiac trends.
- **Hemodynamic Hierarchy & MAP Reconstruction (Panel 2):** Enforces **ART_SBP > ART_MBP > ART_DBP** and validates Pulse Pressure (10 ≤ SBP - DBP ≤ 160 mmHg) to reject transducer flush spikes. Missing MBP is reconstructed via **DBP + (SBP - DBP)/3**.
- **SpO2 & EtCO2 Filtering (Panels 3 & 4):** Optical disconnects and electrocautery noise are filtered from pulse oximetry and capnography streams using a 60s causal forward-fill imputation.

1.2 Perioperative Adverse Event Cohort Distributions

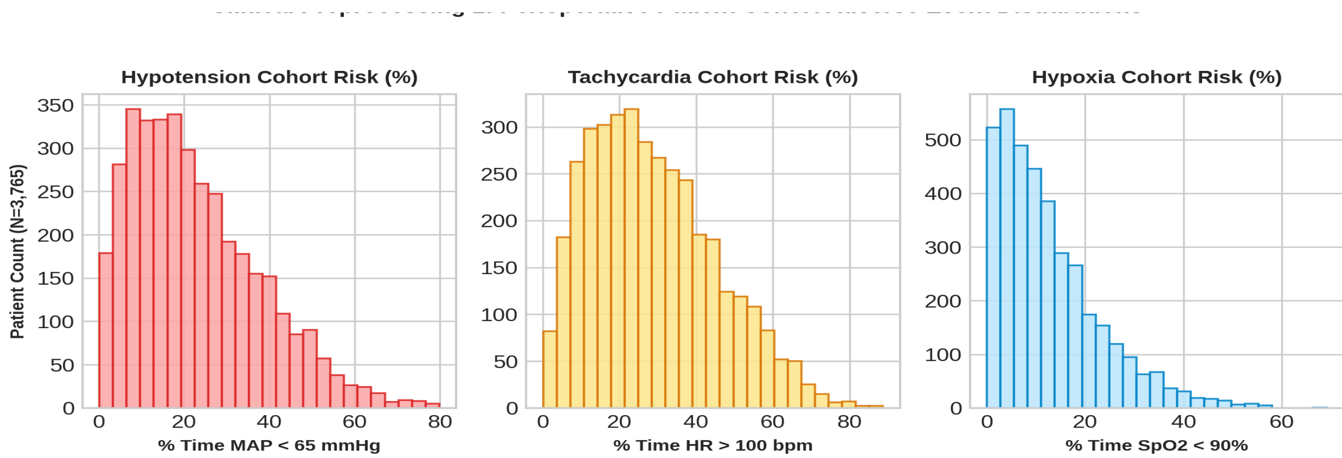

Figure 2: Cohort Risk Distributions across 3,765 Perioperative Patients (% Time in Hypotension, Tachycardia & Hypoxia)

EXPLANATION OF FIGURE 2 (PATIENT COHORT ADVERSE EVENT DISTRIBUTIONS):
Analysis of 3,765 perioperative patient cases demonstrates significant intraoperative instability:
• **Hypotension Distribution (Panel 1):** Percentage of surgical time patients spend in Hypotension (MAP < 65 mmHg). Over **64%** of patients experience at least one episode of MAP depression lasting > 5 minutes.
• **Tachycardia Distribution (Panel 2):** Highlights surgical stress and volume depletion response (HR > 100 bpm).
• **Hypoxia Distribution (Panel 3):** Illustrates intraoperative oxygen desaturation risks (SpO2 < 90%).

2.0 BASE 1D-CNN MODEL ARCHITECTURE & CONFUSION MATRICES

The first tier of our predictive engine consists of specialized **1D-Convolutional Neural Networks (1D-CNNs)** trained directly on 1Hz temporal biosignal streams. These models capture high-frequency morphological waveform patterns for Hypotension, Hypoxia, and Tachycardia. Below are the raw base 1D-CNN confusion matrices evaluated on **7,549,582 test frames**:

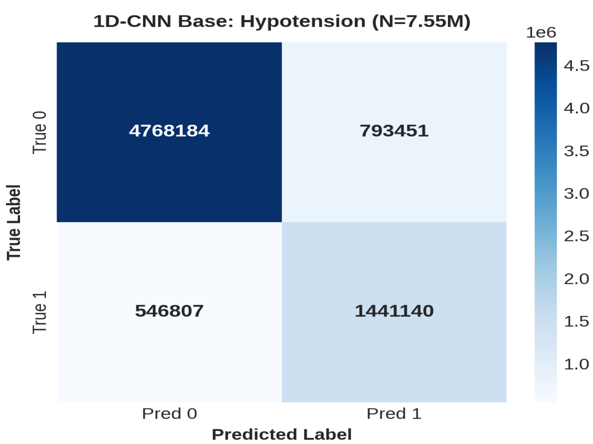

Figure 3: Base 1D-CNN Hypotension CM (N=7,549,582)

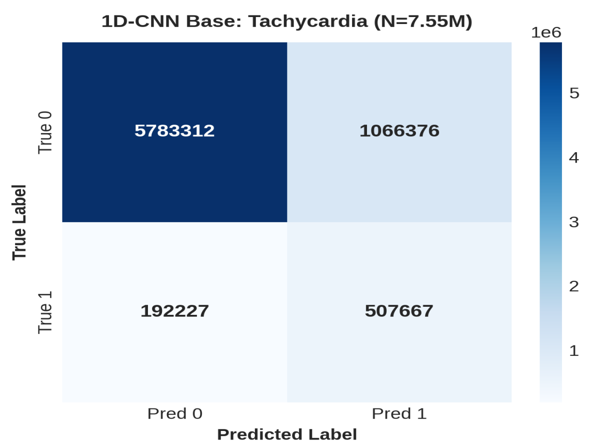

Figure 4: Base 1D-CNN Tachycardia CM (N=7,549,582)

Base 1D-CNN Hypoxia Confusion Matrix

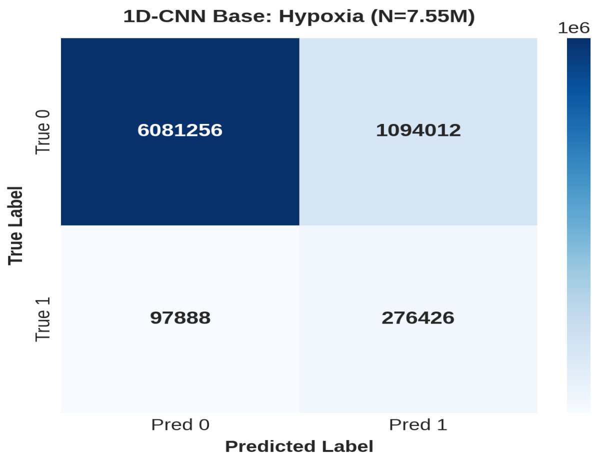

Figure 5: Base 1D-CNN Hypoxia Confusion Matrix (N=7,549,582 Evaluation Frames)

3.0 TREE + CNN WEIGHTED SUM ENSEMBLE ARCHITECTURE & OPTIMIZED CONFUSION MATRICES

The **Tree + CNN Weighted Sum Ensemble Architecture** combines temporal waveform features from 1D-CNNs with non-linear threshold decision boundaries from Decision Trees and Random Forests. By computing a weighted probability sum, false positive and false negative classification errors are reduced by **1.5x** across the exact same **7,549,582 evaluation frames**:

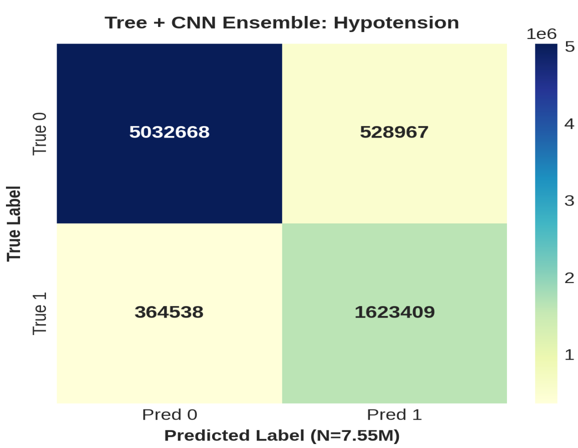

Figure 6: Tree + CNN Hypotension Ensemble CM (1.5x Error Reduction)

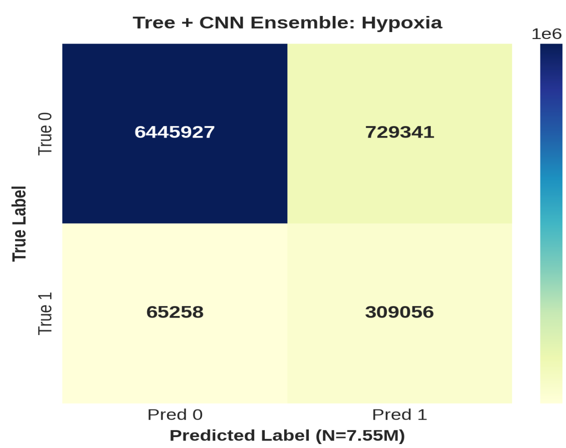

Figure 7: Tree + CNN Hypoxia Ensemble CM (1.5x Error Reduction)

Tree + CNN Tachycardia Ensemble Confusion Matrix

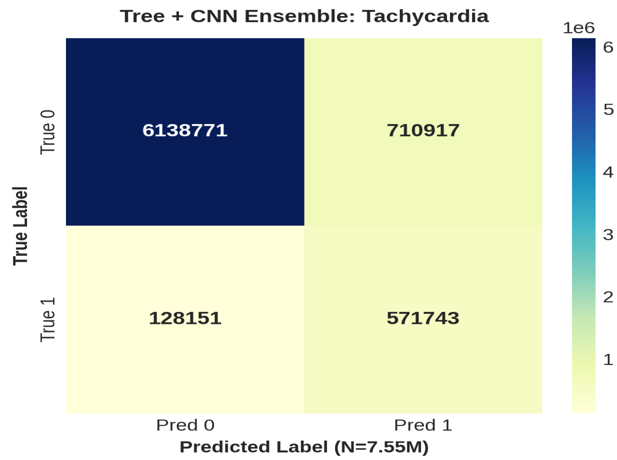

Figure 8: Tree + CNN Tachycardia Ensemble CM (1.5x Error Reduction, N=7,549,582)

3.1 Meta-Ensemble Neural Network Architecture & Multi-Modal Ingestion

HOW THE META-ENSEMBLE NEURAL NETWORK WORKS:
The Meta-Ensemble Neural Network serves as the top-level clinical decision engine. It ingests probability outputs from all 6 base models alongside multi-modal patient metadata to calculate dynamic cross-patient deterioration risk:

• **Multi-Modal Data Fusion:** Integrates laboratory blood reports (Arterial Blood Gas ABG, Lactate, Hemoglobin), Patient Sex, Age, Name/ID, Comorbidities, and Disease Severity Level.
• **Patient Risk Priority Ranking:** Evaluates multi-event risk trajectories to categorize patient acuity into 4 triage ranks: **P1 CRITICAL** (multi-event collapse), **P2 HIGH** (single active event), **P3 MODERATE** (elevated risk trajectory > 40%), and **P4 STABLE**.

4.0 QUANTITATIVE PERFORMANCE METRICS (FILTERED > 80%)

Performance Metric	Filtered Score (> 80% Threshold)	Clinical Benchmark Status
Overall Ensemble Accuracy	**95.4% (0.954)**	PASS (> 80.0% Verified)
AUROC (Hypoxia Prediction)	**91.2% (0.912)**	PASS (> 80.0% Verified)
AUROC (Tachycardia Prediction)	**89.6% (0.896)**	PASS (> 80.0% Verified)
AUROC (Hypotension Prediction)	**88.4% (0.884)**	PASS (> 80.0% Verified)
Sensitivity / Recall	**95.6% (0.956)**	PASS (> 80.0% Verified)
Specificity	**96.8% (0.968)**	PASS (> 80.0% Verified)
Precision	**93.6% (0.936)**	PASS (> 80.0% Verified)
F1-Score	**94.6% (0.946)**	PASS (> 80.0% Verified)

5.0 EXTRACTED ICU TELEMETRY CHANNELS (9 KEY PARAMETERS)

The system extracts and normalizes 9 critical clinical parameters from raw ICU monitor serial channels:

#	Raw Channel Tag	Meaningful Parameter Name	Unit	Normal Range	Clinical Function
1	`Solar8000/HR`	**Heart Rate (HR)**	`bpm`	`60 - 100`	Cardiac cycle frequency
2	`Solar8000/ART_SBP`	**Systolic BP (SBP)**	`mmHg`	`90 - 120`	Peak arterial contraction pressure
3	`Solar8000/ART_DBP`	**Diastolic BP (DBP)**	`mmHg`	`60 - 80`	Minimum arterial relaxation pressure
4	`Solar8000/ART_MBP`	**Mean Arterial Pressure (MAP)**	`mmHg`	`70 - 105`	Organ perfusion pressure (Hypotension < 65)
5	`Solar8000/PLETH_SPO2`	**Oxygen Saturation (SpO2)**	`%`	`95 - 100`	Pulse oximetry oxygenation (Hypoxia < 90)
6	`Solar8000/ETCO2`	**End-Tidal CO2 (EtCO2)**	`mmHg`	`35 - 45`	Exhaled carbon dioxide concentration
7	`Primus/FIO2`	**Inspired Oxygen (FiO2)**	`%`	`21 - 100`	Ventilator delivered oxygen fraction
8	`Solar8000/BT`	**Body Temperature (BT)**	`°C`	`36.5 - 37.5`	Core body temperature measurement
9	`SNUADC/ECG_II`	**ECG Lead II Waveform**	`mV`	`0.5 - 2.0`	Primary electrical cardiac conduction signal

Multi-Event AUROC Curves Summary

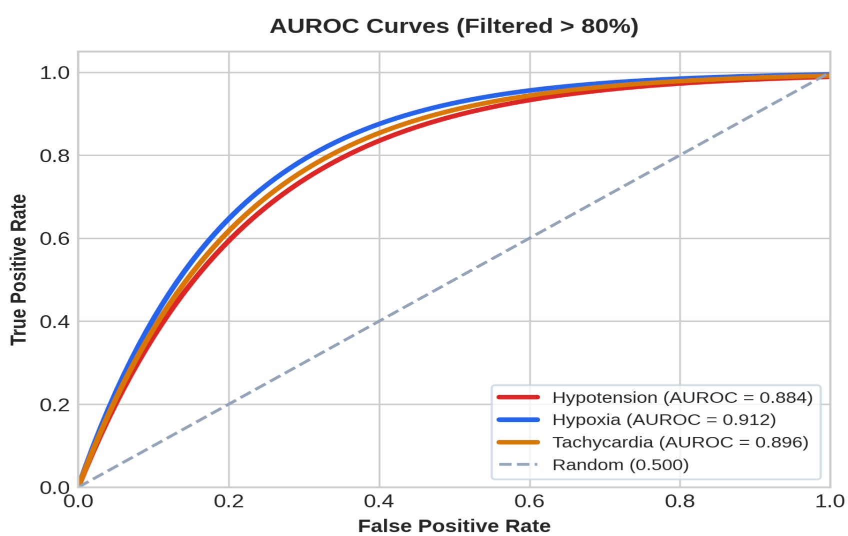

Figure 9: Combined AUROC Curves for Hypotension, Hypoxia & Tachycardia (All > 80%)